Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Solving by Factoring and Other Methods **Date:** ___________

Advanced (Habanero)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.
(A) $x = -2, x = -3$
(B) $x = -1, x = -6$
(C) $x = 2, x = 3$
(D) $x = 1, x = 6$
(E) $x = -2, x = 3$

Q2. Solve the equation $x^2 - 7x + 12 = 0$ by factoring.
(A) $x = 3, x = 4$
(B) $x = 2, x = 6$
(C) $x = 1, x = 12$
(D) $x = 5, x = 12$
(E) $x = 3, x = 12$

Q3. Solve the equation $x^2 + 2x - 6 = 0$ by factoring.
(A) $x = -1pmsqrt7$
(B) $x = 1pmsqrt7$
(C) $x = -3, x = 2$
(D) $x = -2, x = 3$
(E) $x = 1, x = 6$

Q4. Solve the equation $x^2 - 4x - 5 = 0$ by factoring.
(A) $x = -1, x = 5$
(B) $x = 1, x = -5$
(C) $x = -1, x = -5$
(D) $x = 5, x = -1$
(E) $x = -5, x = 1$

Q5. Verify the solutions to the equation $x^2 + 2x - 3 = 0$.
(A) $x = -3, x = 1$
(B) $x = -1, x = 3$
(C) $x = 1, x = -3$
(D) $x = -3, x = -1$
(E) $x = 3, x = 1$

Q6. Solve the equation $x^2 - 9x + 20 = 0$ by factoring.
(A) $x = 4, x = 5$
(B) $x = 5, x = 4$
(C) $x = 1, x = 20$
(D) $x = 9, x = 2$
(E) $x = 4, x = 20$

Q7. Solve the equation $x^2 + x - 12 = 0$ by factoring.
(A) $x = -4, x = 3$
(B) $x = -3, x = 4$
(C) $x = 1, x = 12$
(D) $x = 4, x = 3$
(E) $x = -3, x = -4$

Q8. Solve the equation $x^2 - 2x - 15 = 0$ by factoring.
(A) $x = -3, x = 5$
(B) $x = 3, x = -5$
(C) $x = -5, x = 3$
(D) $x = 1, x = 15$
(E) $x = 5, x = -3$

Open Ended Questions (FRQ)

Q9. Solve the equation $x^2 + 4x + 4 = 0$ by factoring and verify the solutions.

Q10. Solve the equation $x^2 - 5x - 6 = 0$ by factoring and verify the solutions.

Chef's Tips

- Check the calculations carefully.
- Make sure the factoring is correct.
- Verify the solutions by plugging them back into the original equation.